

8.3.3 Reclassification

After each time series cluster had been examined, each individual graph was evaluated and reclassified by how each graph of the clusters matched the understanding of gentrification from the literature review. Each graph was also compared to both municipal and national means from the Danish Neighborhood Atlas to see the changes in a broader perspective.

This led to the conclusion that a big change in e.g. unemployment rate meant that a new demographic composition had occurred and therefore gentrification presumably has taken place at some stage. This can be seen on figure 8.4b which shows unemployment rate divided into six classes.

Meanwhile the areas with a smaller decrease in unemployment rate shows that the demographic composition was stable through this time period from 1990 to 2020. This meant in practice that the brown graph on figure 8.4 was attributed the new value 1 and the yellow graph got value 6. This process was repeated for each dataset linking the changes in the data to how gentrification is measured and seen in practice². See appendix A for further insight.

8.3.4 Overlay Analysis

The aim of the analysis is to show areas that have been subject to gentrification and therefore a map is needed that encapsulates all of these different developments. With a weighted overlay analysis each dataset gets taken into account and weighed by how well it can be used as a measurement of gentrification (ESRI, n.d.-c).

As explained in the literature review in chapter 3, the concept of gentrification is a complex phenomenon with several different definitions and criteria. In the case of Vidal (2019), Drew (2018) and Reades et al. (2019) only a few variables are used as measurements (Square-meter price, disposable income, higher education and unemployment), whereas Institute on Metropolitan Opportunity (2016), Yoo (2023) and Knorr (2019) supplements with several other variables (family composition, age, ethnicity, migration rate etc.).

Because of the different understanding on which data to be used and how, it is decided that the overlay analysis should consist of the DNA data and be weighted on the basis of an 'analytic hierarchy process' (AHP). AHP is a multi-criteria decision method used by

²The reclassification of gentrification-values can be found in the GitHub repository under data/interim/reclass_tsc.xlsx

decision makers when assigning weights to complex problems with many factors.

The weights are determined by having participants do a pairwise comparison of two datasets, which ignores the noise of other dataset in the single decision. This is then iterated for every pairwise comparison, which makes it possible to rank the impact of each dataset (Sajid, 2011, p. 31). An AHP-model called AHP-OS has been used to perform the pairwise comparison. AHP-OS is a web-based online software that uses these methods (Goepel, 2018).

AHP Scale: 1- Equal Importance, 3- Moderate importance, 5- Strong importance, 7- Very strong importance, 9- Extreme importance (2,4,6,8 values in-between).

With respect to **Gentrification**, which criterion is more important, and how much more on a scale 1 to 9?

A - wrt Gentrification - or B?		Equal	How much more?
1	☒ Housing prices	☐ Price pr/sq meter	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9
2	☒ Housing prices	☐ Public housing	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9
3	☒ Housing prices	☐ Inflow of residents	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9
4	☒ Housing prices	☐ Outflow of residents	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9
5	☒ Housing prices	☐ Couples with children	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

Figure 8.5: AHP decision table for the first 5 questions (Goepel, 2018))

By distributing an AHP-project to the group participants as well as selected members of the Planning Faculty of Aalborg University, it is sought to get a broad understanding of both the similar and the contrasting views on gentrification. The weighing is done based on the background knowledge that the group participants and faculty members possess of the subject. Each participant was asked, how much the given parameter influence the gentrification process and these questions can be seen on figure 8.5. Some of the measuring and quantifying in past studies regarding gentrification is not done transparently (see section 3.4), this method is therefore more robust than simply assigning random weights to datasets. The weights are set as shown in table 8.1

Table 8.1: Weights assigned via AHP distribution by six participants, P1–P6 [%]

Layer	P1	P2	P3	P4	P5	P6	Result
Disposable income	6.4	17.3	11.2	19.2	4.9	13.3	12.1
Public housing	11.7	1.3	13.6	1.8	23.6	6.3	9.7
Price per m ²	7.6	13.5	5.1	11.0	10.6	5.5	8.9
Housing prices	9.4	15.8	2.1	9.1	9.2	7.1	8.8
Outflow of residents	10.9	2.9	4.0	1.3	14.0	9.6	7.1
Unemployed	6.3	8.5	2.3	8.7	2.1	12.8	6.8
Higher education	1.5	7.4	11.6	13.8	2.7	1.3	6.4
Employed	6.7	5.3	8.7	1.4	2.1	12.8	6.2
Inflow of residents	8.2	2.6	3.8	1.3	9.7	9.2	5.8
Outside the labor market	6.3	8.6	8.9	1.1	2.1	2.7	5.0
26–40 years	2.7	1.1	8.7	5.7	1.3	2.2	3.7
Primary and lower second. schl.	1.8	2.7	1.1	4.6	1.9	8.7	3.5
Upp. second. schl. and voc. coll.	1.6	1.9	1.7	7.8	1.9	1.3	2.7
56–69 years	2.7	0.8	1.6	1.4	1.8	0.7	2.6
Serious crime	2.7	4.4	0.7	2.1	1.9	2.3	2.4
Couples without children	2.7	2.2	3.4	2.1	1.8	1.0	2.2
41–55 years	2.7	0.8	5.7	1.3	2.2	0.7	2.2
Couples with children	2.7	1.3	3.2	2.4	1.9	0.9	2.1
18–25 years	2.7	0.8	1.3	2.9	2.0	0.7	1.7
Singles with and without children	2.6	0.7	1.6	0.8	1.8	1.0	1.4

The result of the AHP-weights are then applied to an overlay analysis where clusters with a lower overall value means that it is considered more gentrified than a neighborhood with a higher value. The lowest value are 1.75 and the highest are 4.86, with a standard deviation of 0.54 units. The overlay map is converted from a raster image to a shapelayer with the tool 'zonal statistic'.

The result is shown on on figure 8.6. The map shows the result of the weighted overlay analysis and group each neighborhood by the extent of possible gentrification based on the evolution of each dataset at that exact location. As anticipated both Inner Vesterbro and Nørrebro show clear signs of having gone through gentrification as both areas mostly consist of red and orange neighborhoods. Areas in Østerbro (Aarhusgade kvarteret) and Amager (Amagerbro) also show signs of gentrification. This is not yet areas that are commented on in the literature, however both areas have seen a large increase in housing prices and residents with higher income and education level.

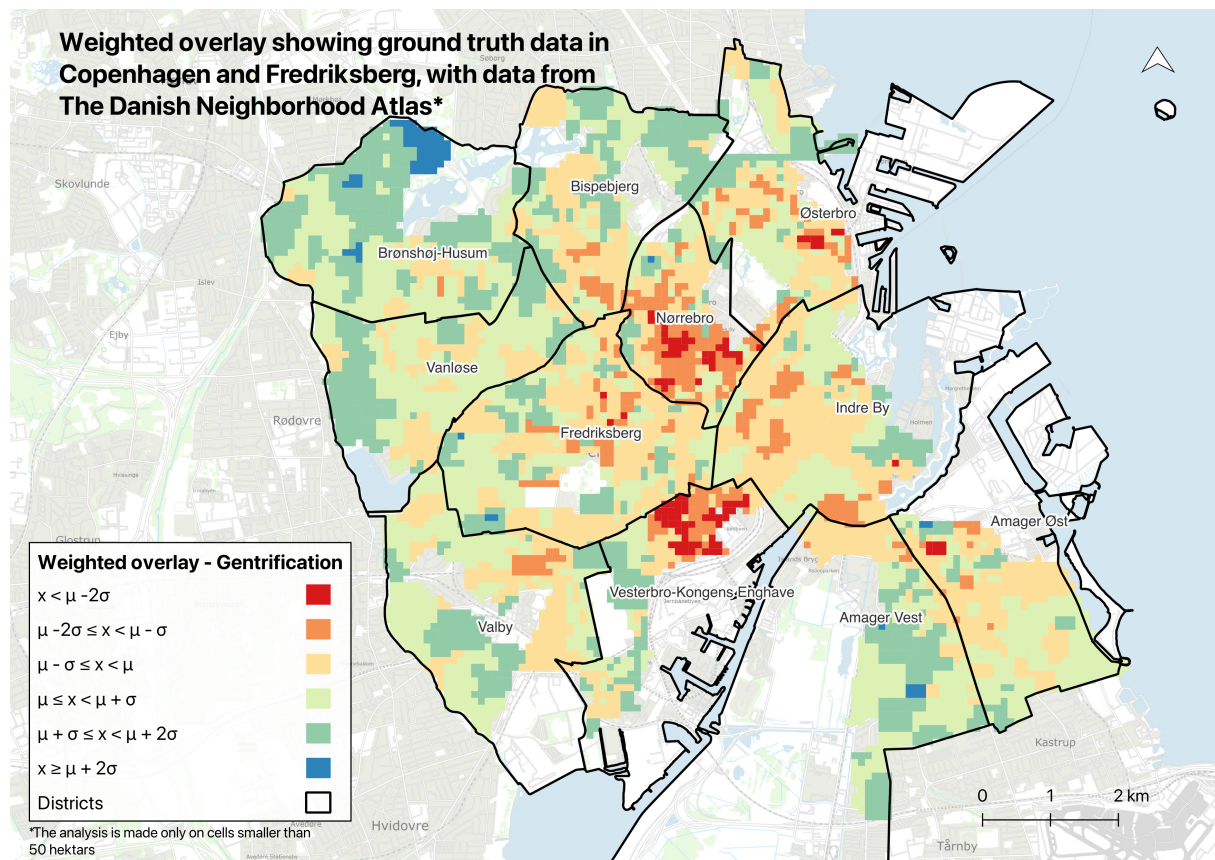


Figure 8.6: Gentrified areas in Copenhagen and Frederiksberg Municipality

Before this GTD can function in the ML model, it needs to be converted from discrete values to integer values. This is done firstly with a standard deviation classification as shown on table 8.2. The ML model was initially supposed to run with six classifications, however this produced very low accuracy within the first test-runs. It was then decided upon a 3-class scale which can be seen on the table below.

Table 8.2: Table showing the gentrification values from a discrete number to integer classification

Discrete value	Cluster count	Standard deviation	Initial classification (1 to 6)	Final classification (0 to 2)
[1.757; 2.301]	52	$x < \mu - 2\sigma$	1	0
[2.301 ; 2.845]	152	$\mu - 2\sigma \leq x < \mu - \sigma$	2	
[2.845 ; 3.389]	519	$\mu - \sigma \leq x < \mu$	3	1
[3.389 ; 3.933]	424	$\mu \leq x < \mu + \sigma$	4	
[3.933 ; 4.477]	265	$\mu + \sigma \leq x < \mu + 2\sigma$	5	2
[4.477 ; 5.021]	9	$x \geq \mu + 2\sigma$	6	

The three classes in the GTD ³ consist of: 0: Likely to be gentrified, 1: No trends, 2: Unlikely to be gentrified. The ground truth data are ready to be applied on the training data for Copenhagen and Frederiksberg and a map of the final classifications can be seen below on figure 8.7.

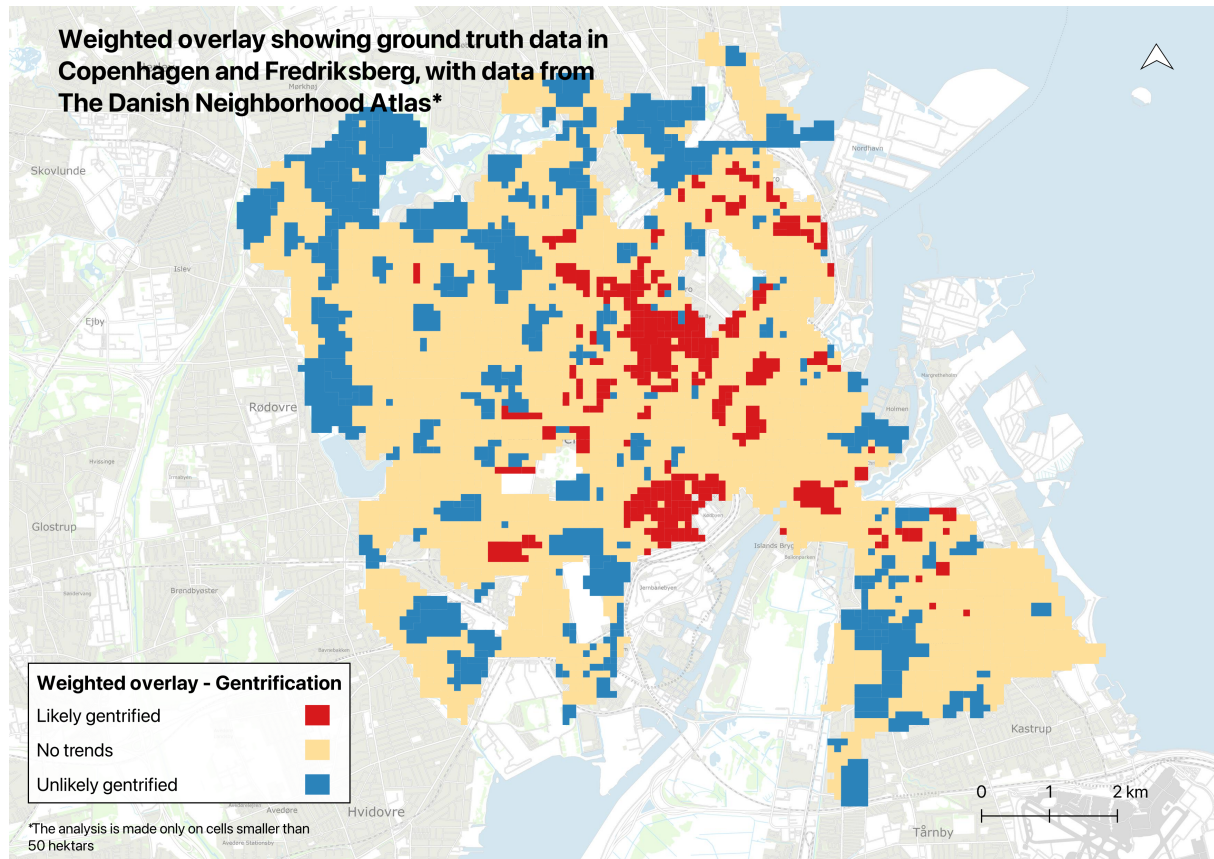


Figure 8.7: Ground truth data divided in three classes

The map on figure 8.7 shows many areas as red and not just in Inner Vesterbro and Nørrebro like the studies in the literature review pointed out. This is likely due to the fact that gentrification is not only happening in these areas, but can also happen in a smaller scale, as pointed out on page 13. Patterns can still be deduced from the map as the areas classified as likely to be gentrified (red) are mostly located in the central part especially Vesterbro and Nørrebro but also Indre By, Østerbro, Christianshavn and Amagerbro. Most of the neighborhoods are classified as no trends (yellow) meaning the degree of gentrification is neutral or unclear. The areas unlikely to be gentrified are mostly in western edge of Copenhagen and Frederiksberg where the model found no or very few signs on gentrification.

³The reason why the classes are 0,1,2 is because of python starts lists with 0

9 | Machine Learning Model

The following chapter will focus on how the machine learning model has been developed and tuned to be as strong as possible in predicting gentrified neighborhoods in 2020. The project file for the model can be found in the GitHub repository. The folder contains all notebooks regarding the model and dummy files for the model and ground truth data. The README.txt contains all the necessary information about how the project folder is structured and how to run the model on a local machine.

9.1 Model Selection

During the development of the ML model there has been several versions of different models. As a baseline, an earlier paper about predicting landslides was used as inspiration on which models to use (Ageenko et al., 2022). As a first step three different models were tested. This was the same process as in Ageenko et al. (2022) and the models tested were random forest classification, linear regression and a vector machine. These models were useful in Ageenko et al. (2022) due to the accurate nature of scientific data related to landslides but these models cannot process no data values. Since the chosen dataset from DNA contains a lot of no data values it was decided not to work with any of these models in the ML model.

It was decided to work with the model XGBoost (eXtreme Gradient Boosting). XGBoost is a newly developed tree based model that can handle no data values. It is also one of the most powerful tree based models for supervised machine learning and is especially good at handling complex tabular data (DMLC XGBoost, 2025).

9.2 Model Workflow

In this section the model workflow is outlined consisting of how the process went from raw data from DNA to a final prediction of all neighborhoods under 50ha. The training area consists of the municipalities of Copenhagen and Frederiksberg and the test area is all the neighborhoods in the 34 municipalities of the Finger Plan from 1947 under 50ha. The Finger Plan is chosen as a study area as it is composed of both mid-sized towns as